

PROJECT 2

DECISION THEORY AND MULTI-CRITERIA ANALYSIS

Alessia Aquilini, Gabriele Battagliola, Luigina Bertolotti, Federico Buizza, Marie Doka, Samuela Paci, Martina Pagan

PROBLEM DESCRIPTION

The Italian oil mill company Viola has to plan, starting from the production plant located in Venosa (il), the delivery of extra virgin oil to its customers located in 10 cities in the South of Italy. The company uses for the deliveries a fleet of 3 identical vehicles having a capacity equal to 3500 liters. The delivery takes place once a month and the production capacity of the company is 8000 liters per month.

At the beginning of the month the price of olive oil is 45,2€/l, whether in the second period we consider 3 different scenarios with π_w probability. In the first scenario, which is the only one considering a depreciation, the olive oil price is 43,5€/l, in the second one it is 46,4€/l and in the third one it is 48,2€/l.

Our objective is to maximize the company's profit, considering the uncertain variable price, the distribution and production costs.

The distance between each pair of cities, expressed in kilometers, represents the cost of transporting the olive oil from one city to another.

CVRP

Capacitated Vehicle Routing Problem

Before implementing the two-stage stochastic problem, we focused on determining optimal routes for vehicles, with the aim of minimizing the total travel distance.

SETS:

- **I**: set of all nodes (depot $i=1$ and customers $i \in \{2, \dots, I\}$).
- **k**: set of vehicles $k=1, \dots, 3$
- **i,j** $\in I$: nodes in the network.

PARAMETERS:

- d_{ij} : distance between nodes i and j .
- q_i : demand at customer i (depot has no demand).
- **B**: vehicle capacity.

DECISION VARIABLES:

- $x_{ijk} \in \{0, 1\}$: binary variable indicating whether vehicle k travels from node i to node j .
- $P_{ik} \geq 0$: auxiliary variables for subtour elimination, Miller-Tucker-Zemlin (MTZ) method.

OBJECTIVE FUNCTION

$$\min \sum_{k \in K} \sum_{i \in I} \sum_{j \in I, i \neq j} d_{ij} * x_{ijk}$$

Subject to

- Flow Conservation:

$$\sum_{i \in I, i \neq j} x_{ijk} = \sum_{i \in I, i \neq j} x_{jik}, \forall j \in I \setminus \{1\}, k \in K$$

- Visit Each Customer Exactly Once:

$$\sum_{k \in K} \sum_{i \in I, i \neq j} x_{ijk} = 1, \forall j \in I \setminus \{1\}$$

- Depot Departure:

$$\sum_{j \in I \setminus \{1\}} x_{1jk} = 1, \forall k \in K$$

- Vehicle Capacity:

$$\sum_{i \in I \setminus \{1\}} \sum_{i \in I, i \neq j} q_i * x_{ijk} \leq B, \forall k \in K$$

- Subtour Elimination (MTZ):

$$P_{ik} - P_{jk} + |I|x_{ijk} \leq |I| - 1, \forall i, j \in I \setminus \{1\}, i \neq j, k \in K$$

$$P_{ik} \geq 0, P_{1k} = 0, P_{ik} \leq |I| - 1, \forall i \in I, k \in K$$

TWO-STAGE STOCHASTIC PROBLEM

INDEX:

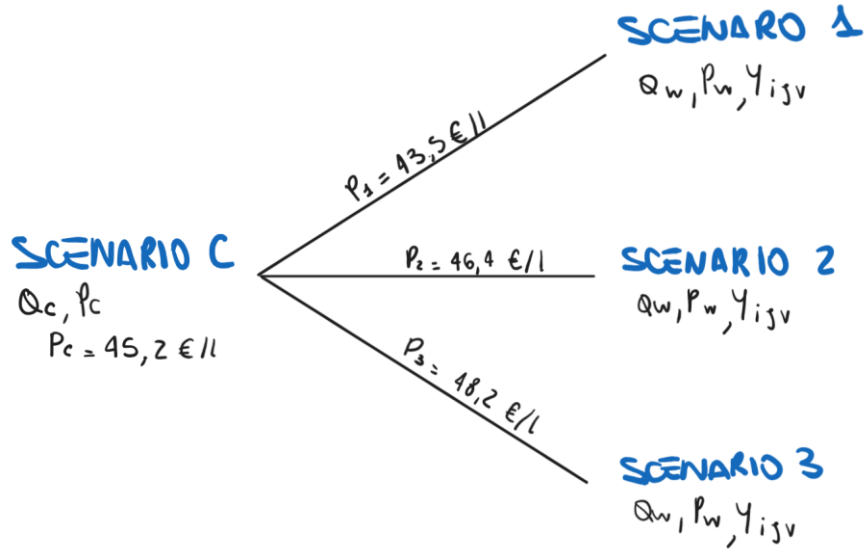
- **i**: cities, starting point $i=1, \dots, 11$
- **j**: cities, delivering point $j=1, \dots, 11$
- **v**: vehicles $v=1, \dots, 3$
- **w**: scenarios $w=1, \dots, 3$

PARAMETERS:

- **C**: vehicles capacity, 3500 liters for each vehicle
- **D_i** : demand for cities i
- **c_{ij}** : cost from city i to city j
- **k** : production capacity, 8000 liters for month
- **β** : beta $\beta = 0, \dots, 1$
- **p_w** : price per liter in each scenario
- **π_w** : probability for each scenario
- **f** : cost of the production ($\approx 30\text{€}$)
- **p_c** : price per liter in the first stage

VARIABLES

- **Y_{ijv}** :
 - **$Y_{ijv} = 0$** $\rightarrow$ vehicle not used
 - **$Y_{ijv} = 1$** $\rightarrow$ vehicle used
- **Q_w** : quantity produced for each scenario
- **ε_w** : ksi, variable for the CVaR
- **η** : eta, variable for the CvaR
- **Q_c** : quantity produced in the first stage



NODE-VARIABLE FORMULATION

$$\max_{Q_c, Q_w} (1 - \beta) \left[\left(p_c * Q_c + \sum_{w=1}^3 \pi_w * (p_w * Q_w) \right) - \left(\left(Q_c + \sum_{w=1}^3 \pi_w * Q_w \right) * f + \sum_{i,j} c_{i,j} \right) \right] - \beta \left(\eta - \left(\frac{1}{0.1} \right) * \sum_{w=1}^3 \pi_w * \varepsilon_w \right)$$

Subject to

- Positive variables

$$Q_c, Q_1, Q_2, Q_3 \geq 0$$

$$\varepsilon_w \geq 0$$

- Production capacity

$$Q_c + Q_w \leq k, \forall w$$

- CVaR constraint

$$\eta - (Q_c + \pi_w * Q_w) * f - \sum_{i,j} c_{ij} \leq \varepsilon_w, \forall w$$

- Production demand constraint

$$Q_c + Q_w \geq \sum_i D_i, \forall w$$

- Route for vehicle 1

$$y_{17}^{v1} = 1, y_{76}^{v1} = 1, y_{65}^{v1} = 1, y_{54}^{v1} = 1, y_{41}^{v1} = 1$$

- Route for vehicle 2

$$y_{12}^{v2} = 1, y_{211}^{v2} = 1, y_{1110}^{v2} = 1, y_{109}^{v2} = 1, y_{98}^{v2} = 1, y_{81}^{v2} = 1$$

- Route for vehicle 3

$$y_{13}^{v3} = 1, y_{31}^{v3} = 1$$